



The AI Case Worker for Refugees, NGOs, and Vulnerable Communities

“Explain. Translate. Verify. Guide.”

AidBridge is a **human-centered AI system** that helps people who are:

- Refugees
- Migrants
- Low-income families
- Illiterate or semi-literate

understand:

- Legal documents
- Aid eligibility
- Medical, housing, and school forms

and helps **NGOs process cases faster and safer.**

1 Real-World Problem

Refugees and low-income families face:

- Forms in legal English or formal Arabic
- Scanned documents
- Conflicting rules from multiple NGOs
- No lawyers or translators

Mistakes mean:

- Rejected applications
- Lost healthcare
- No housing
- Deportation risk

This is not theoretical.

UNHCR, IRC, NRC, and local NGOs struggle with this every day.

2 What AidBridge Does

AidBridge is both:

- A **self-service assistant** for people
- A **case-management assistant** for NGOs

People can:

- Upload papers
- Speak in Arabic
- Ask “What does this mean?”
- Ask “Am I eligible?”

NGOs can:

- Verify information
 - See extracted facts
 - Approve or reject
 - Audit decisions
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3 MVP Scope (4 weeks)

Feature	Included
Arabic & English	✓
Voice input	✓
OCR for documents	✓
RAG over laws & NGO rules	✓
Case creation	✓

Feature	Included
Multi-agent verification	✓
PII protection	✓
Audit trail	✓
NGO dashboard	✓

4 System Architecture

Mobile-friendly Web App

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Backend API

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| Users | Cases | Docs | Logs | NGO |

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RAG Engine

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Vector DB (Pinecone / pgvector)

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Agent Orchestrator

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OpenAI + Arabic models (Jais/AraGPT)

5 Multimodal AI

AidBridge uses **3 modalities** (Gold-tier):

Mode	Purpose
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Vision	Read scanned IDs, forms
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Speech	Refugees speak, not type
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Text	NGO staff and records
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RAG System (Truth over hallucination)

Documents stored:

- Immigration law
- NGO eligibility rules
- Aid programs
- Medical and housing policies

When AI answers:

“Are you eligible for housing?”

It must:

- Retrieve law
- Retrieve NGO rules
- Cite both

No free-text guessing.

Multi-Agent System

AidBridge uses **Sequential Agents**:

Intake Agent → Eligibility Agent → Verification Agent → Case Summary Agent

Agent	Role
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Intake	Understand user story
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Agent	Role
Eligibility	Check laws + NGO
Verification	Detect fraud / missing
Summary	Produce NGO report

8 Tools

Tool	Use
ocr_doc()	Extract text
check_law()	RAG
check_ngo_rules()	RAG
store_case()	DB
redact_pii()	Safety
log_audit()	Compliance

9 Safety & Ethics

This project has **the strongest security of all three.**

Risk	Control
PII leaks	Auto-redaction
Surveillance misuse	Consent + access logs
Bias	Dual-agent verification
Hallucinations	RAG-only mode
Legal harm	Citation required

1 0 Why This Is Real

This matches:

- UNHCR Digital Transformation
- IRC case management
- EU refugee support systems
- Civic-tech startups

This is the kind of AI NGOs actually want — but safer than chatbots.

1 1 What students learn

They will implement:

- Voice AI
- OCR
- Multilingual RAG
- Ethical AI
- PII handling
- Audit logs
- Agent verification

This is **world-class portfolio material**.